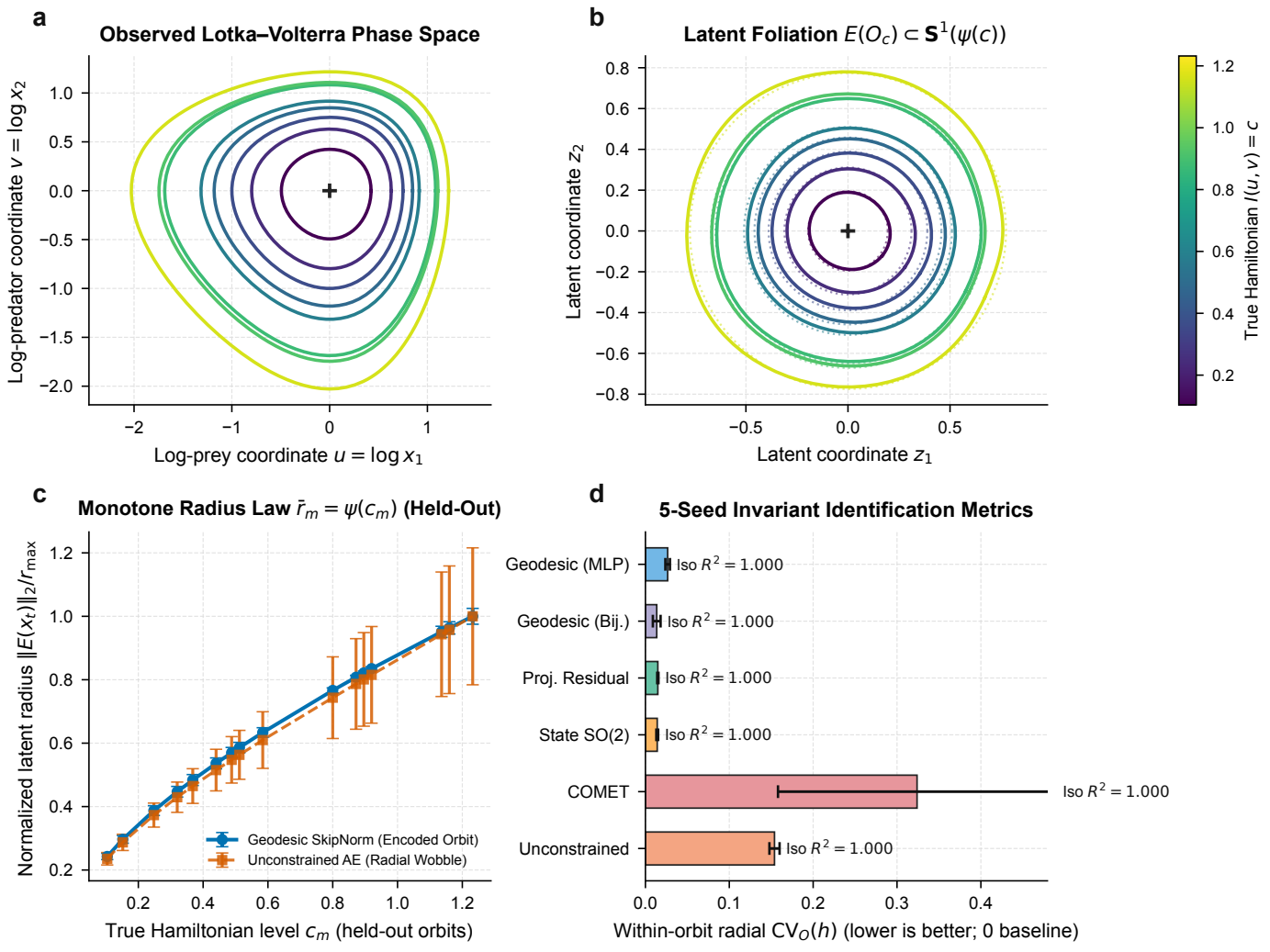

a
Observed Lotka–Volterra Phase Space
Log-predator coordinate $v = \log x_2$
Log-prey coordinate $u = \log x_1$
b
Latent Foliation $E(O_c) \subset \mathbf{S}^1(\psi(c))$
Latent coordinate z_2
Latent coordinate z_1
True Hamiltonian $I(u, v) = c$
c
Monotone Radius Law $\bar{r}_m = \psi(c_m)$ (Held-Out)
Normalized latent radius $\|E(x_t)\|_2 / r_{\max}$
True Hamiltonian level c_m (held-out orbits)
Geodesic SkipNorm (Encoded Orbit)
Unconstrained AE (Radial Wobble)
d
5-Seed Invariant Identification Metrics
Geodesic (MLP)
Geodesic (Bij.)
Proj. Residual
State SO(2)
COMET
Unconstrained
Iso $R^2 = 1.000$
Within-orbit radial $\mathrm{CV}_O(h)$ (lower is better; 0 baseline)